Movement into and out of cells

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Diffusion [1]

Which statement describes diffusion?

A the movement of particles from a region of lower concentration to a region of higher concentration, using energy from respiration

B the net movement of particles from a region of higher concentration to a region of lower concentration, as a result of their random movement

C the movement of water molecules only, through a partially permeable membrane

D the movement of particles from a region of lower concentration to a region of higher concentration, as a result of their random movement

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2 · Part 1 Core Diffusion [1]

Where does the energy for diffusion come from?

A energy released by respiration in the cell

B energy supplied by protein carriers in the cell membrane

C light energy absorbed by the particles

D kinetic energy of the random movement of the particles

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3 · Part 1 Core Diffusion [1]

Four identical blocks of colourless gelatine are placed in a red dye solution for 10 minutes. Some of the blocks are coated with wax so that only one face is in contact with the dye.

The table shows the conditions for each block.

Which block will have the largest volume coloured red after 10 minutes?

	temperature / °C	concentration of dye solution / %	faces in contact with the dye
A	40	10	all six faces
B	40	2	one face
C	20	10	one face
D	20	2	all six faces

4 · Part 1 Core Osmosis [1]

Which statement about osmosis is correct?

A It is the movement of solute molecules through a partially permeable membrane.

B It needs energy released by respiration.

C It is the diffusion of water through a partially permeable membrane.

D It only happens in plant cells, because they have a cell wall.

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5 · Part 1 Core Osmosis [1]

Dialysis tubing is filled with a concentrated sucrose solution. The tubing is tied at both ends, dried and weighed.

The tubing is placed in a beaker of distilled water for one hour. It is then removed, dried and weighed again.

Which row shows the results?

	mass of tubing and contents	**concentration of sucrose solution inside the tubing**
A	increased	decreased
B	increased	increased
C	decreased	increased
D	decreased	decreased

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6 · Part 1 Core Osmosis [1]

Pieces of potato were weighed and placed in sucrose solutions of different concentrations. After two hours they were weighed again.

The table shows the percentage change in mass of each piece.

concentration of sucrose solution / %	percentage change in mass / %
0	+12
5	+6
10	+1
15	−5
20	−10

Between which two concentrations of sucrose solution would a potato piece show no change in mass?

A 0 % and 5 %

B 5 % and 10 %

C 15 % and 20 %

D 10 % and 15 %

7 · Part 1 Extended Osmosis [1]

Two sucrose solutions, X and Y, are separated by a partially permeable membrane.

Solution X is 5 % sucrose. Solution Y is 20 % sucrose.

Which statement describes the net movement of water molecules?

A from Y to X, because Y has a higher water potential

B from X to Y, because X has a higher water potential

C from Y to X, because Y has a lower water potential

D from X to Y, because X has a lower water potential

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8 · Part 1 Core Osmosis [1]

The soil around a potted plant is left to dry out for several days. The leaves and soft stem of the plant droop (wilt).

Which statement explains why the plant wilts?

A The cell walls become thinner, so they can no longer hold the plant upright.

B The cells take in too much water by osmosis and burst.

C There is less pressure of water inside the cells pressing outwards on the cell walls, so the cells no longer support the plant.

D The cell membranes become fully permeable, so solutes leak out of the cells.

9 · Part 1 Core Active transport [1]

Cells lining the small intestine continue to absorb glucose from the gut even when the concentration of glucose inside the cells is higher than in the gut.

Which row shows the process that moves glucose into these cells and the source of the energy for it?

	process	source of energy
A	diffusion	kinetic energy of glucose molecules
B	diffusion	respiration
C	active transport	kinetic energy of glucose molecules
D	active transport	respiration

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10 · Part 1 Extended Active transport [1]

Cells in the gills of a freshwater fish take up sodium ions from river water. The concentration of sodium ions in the river water is lower than in the gill cells.

When a substance that stops respiration is added to the water, the uptake of sodium ions stops.

Which statement explains this?

A The sodium ions are moved across the membrane by protein carriers that need energy from respiration.

B The sodium ions enter by diffusion, which needs energy from respiration.

C The substance reduces the kinetic energy of the sodium ions, so they stop diffusing into the cells.

D The sodium ions enter by osmosis, which stops when the water potential of the cells changes.

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11 · Part 1 Core Diffusion [6]

Some single-celled organisms live in pond water. Gases move between the pond water and the organism by diffusion.

The table shows the concentrations of two gases in the pond water and inside one of these organisms.

gas	concentration in pond water / arbitrary units	concentration inside the organism / arbitrary units
oxygen	9	2
carbon dioxide	1	6

(a) Describe what is meant by diffusion. [2]

(b)

(i) Using the table, state the direction in which each gas moves by diffusion. [2]

(ii) Name the part of the cell that the gases must pass through to enter or leave the organism. [1]

(c) Suggest why the organism would die if oxygen could no longer diffuse into it. [1]

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12 · Part 1 Core Diffusion [6]

Students investigated the effect of temperature on the rate of diffusion.

They made identical cubes of agar containing a dilute alkali and an indicator. The indicator is pink in alkali and becomes colourless in acid, so the cubes were pink at the start. Each cube was placed in the same volume of dilute hydrochloric acid at a different temperature. The students recorded the time taken for each cube to become completely colourless.

temperature / °C	time to become colourless / s
15	420
25	330
35	260
45	200

(a) Describe the effect of temperature on the time taken for the cubes to become colourless. Use data from the table in your answer. [2]

(b) Explain why the cubes became colourless faster at higher temperatures. [2]

(c) State two variables, other than the volume of acid, that the students should keep constant. [2]

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13 · Part 1 Core Diffusion [7]

Cubes of agar coloured with a dye can be used as models of cells. Three cubes of different sizes were placed in a solution that removes the colour as it diffuses into the agar. The time taken for the colour to disappear from the centre of each cube was recorded.

length of side of cube / cm	surface area / cm^2	volume / cm^3	surface area : volume ratio	time for colour to disappear / min
1	6	1	6 : 1	4
2	P	8	Q	16
4	96	64	1.5 : 1	63

(a) Calculate the surface area, P, of the cube with sides of 2 cm. Show your working. [2]

(b) Calculate the surface area : volume ratio, Q, of the cube with sides of 2 cm. [1]

(c) Use the information in the table to explain why the colour disappeared from the 1 cm cube in the shortest time. [2]

(d) Suggest why a large multicellular organism cannot get enough oxygen to all its cells by diffusion through its outer body surface alone. [2]

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14 · Part 1 Core Osmosis [4]

Water is an important solvent in living organisms.

(a) Describe the role of water as a solvent in digestion. [1]

(b) Describe the role of water as a solvent in excretion. [1]

(c) Describe the role of water as a solvent in transport. [1]

(d) Name the process by which water moves into and out of cells through the cell membrane. [1]

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15 · Part 1 Core Osmosis [6]

Students set up three bags made of dialysis tubing, P, Q and R. Dialysis tubing is partially permeable: water molecules can pass through it but sucrose molecules cannot.

Each bag was filled with a liquid, tied, weighed and placed in a beaker of another liquid. After one hour each bag was removed, dried and weighed again.

bag	liquid inside the bag	liquid in the beaker	starting mass / g	final mass / g
P	distilled water	30 % sucrose solution	12.0	10.2
Q	30 % sucrose solution	distilled water	12.0	14.7
R	30 % sucrose solution	30 % sucrose solution	12.0	12.0

(a) Calculate the percentage change in mass of bag Q. Show your working. [2]

(b) Explain the change in mass of bag Q. [3]

(c) Explain why the mass of bag R did not change. [1]

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16 · Part 1 Extended Osmosis [7]

Two thin strips of epidermis were peeled from a red onion. Strip X was placed in distilled water. Strip Y was placed in a concentrated sodium chloride solution. After 20 minutes, the cells in each strip were observed with a microscope.

strip	observation
X	the vacuole filled most of each cell and the cytoplasm was pressed against the cell wall
Y	the vacuole was much smaller and the cell membrane and cytoplasm had pulled away from the cell wall in many places

(a)

(i) State the term used to describe the cells in strip X. [1]

(ii) State the term used to describe the cells in strip Y. [1]

(b) Explain, in terms of water potential, the appearance of the cells in strip Y. [3]

(c) Strip Y is then moved into distilled water for 20 minutes. Predict and explain what happens to its cells. [2]

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17 · Part 1 Core Osmosis [5]

Strips of carrot, each 50 mm long, were placed in sodium chloride solutions of different concentrations for three hours. The strips were then measured again.

concentration of sodium chloride solution / %	final length / mm
0	53
2	51
4	48
6	46

(a) Calculate the percentage change in length of the strip in the 6 % solution. Show your working. [2]

(b) The strip in the 0 % solution felt firm. The strip in the 6 % solution was soft and bent easily.

Explain why the strip in the 0 % solution was firm. [2]

(c) Use the table to estimate the concentration of sodium chloride solution in which a carrot strip would not change in length. [1]

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18 · Part 1 Extended Osmosis [7]

Root hair cells absorb water from the soil.

(a) Explain how water enters root hair cells from the soil. Use the term water potential in your answer. [3]

(b) In some fields, salt builds up in the soil over many years. Crop plants growing in this soil often wilt, even when the soil is wet.

Suggest why. [2]

(c) Red blood cells placed in distilled water swell and burst. Plant cells placed in distilled water swell but do not burst.

Explain this difference. [2]

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19 · Part 1 Core Active transport [5]

The table shows the concentrations of two ions in seawater and inside the cells of a seaweed.

ion	concentration in seawater / arbitrary units	concentration inside seaweed cells / arbitrary units
potassium	10	400
sodium	470	60

(a) Identify the ion that must be taken into the seaweed cells by active transport. [1]

(b) Explain your answer to (a) using the data in the table. [1]

(c) State the process in the cells that releases the energy needed for active transport. [1]

(d) The seaweed takes up this ion more slowly at 5 °C than at 20 °C.

Suggest why. [2]

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20 · Part 1 Extended Active transport [6]

Root hair cells take up nitrate ions from soil water.

In one soil, the concentration of nitrate ions in the soil water was 0.4 arbitrary units and the concentration inside the root hair cells was 10 arbitrary units.

The rate of nitrate uptake by the roots was measured in well-aerated soil and in waterlogged soil, where the air spaces between soil particles are filled with water.

soil condition	rate of nitrate uptake / arbitrary units
well-aerated	30
waterlogged	6

(a) Calculate how many times greater the nitrate ion concentration is inside the root hair cells than in the soil water. [1]

(b) Explain why nitrate ions cannot be taken into the root hair cells by diffusion in this soil. [1]

(c) State what moves nitrate ions across the cell membrane of the root hair cell during active transport. [1]

(d) Explain why the rate of nitrate uptake was lower in waterlogged soil. [2]

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(e) Suggest why a plant that could not carry out active transport in its root hair cells would grow poorly in this soil. [1]

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